Tidal potential measured on a coastal–inland optical-clock baseline

Abstract — draft 4

The Moon and Sun change Earth's gravitational potential through their tidal fields and the response of the solid Earth and oceans. Advances in optical clocks and fibre networks have made clocks precise probes of spacetime, providing an experimental foundation for the longstanding vision of monitoring Earth's gravitational potential with a global clock network. Tidal potential changes alter the relative ticking rates of clocks at different locations. Here we detect tidal variations in the potential difference between coastal Shanghai and inland Wuhan using strontium and ytterbium clocks 692 km apart. The predicted principal lunar signal has a fractional-frequency amplitude of 4.78×10^{-18}, equivalent to a height difference of about 4.4 cm. From 545.9 hours of observations spanning 353 days, we find an overall tidal response of 0.96 ± 0.11 relative to the combined solid-Earth and ocean-loading prediction (one statistical standard error). The measured spectrum shows distinct peaks at 22.361 and 23.147 μHz, near the principal lunar M_2 and solar S_2 frequencies, respectively. At these frequencies, the observed spectral power is 46.6 and 8.55 times that remaining after subtraction of the tidal prediction. These observations show that optical clocks can sense the changing gravitational environment over hundreds of kilometres and distinguish neighbouring tidal rhythms within it.